API REFERENCE • VERSION 1.0 (PHASES 1 TO 3)

Taggiot Handheld API

POST-only REST API of the Site Gateway for handheld RFID readers: operator login, finder, binding, inventory count and stock opname.

Taggiot RFID Gate Platform • Site Gateway API

Base URL: https://<site-host>/api/v1

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Overview

Every endpoint in this document uses **HTTP POST** with a JSON body. Identifiers, filters, cursors and limits travel in the body, never in the URL or query string.

Authentication	5 endpoints
Finder	4 endpoints
Binding and editing EPC	2 endpoints
Inventory count and stock opname	11 endpoints
Site admin: operators	4 endpoints
Site admin: handhelds	4 endpoints

Conventions

Topic	Rule
Base URL	`https://<site-host>/api/v1` (the customer's Site Gateway, HTTPS only)
Method and body	POST, `Content-Type: application/json`, UTF-8. Field names are `snake_case`.
Success codes	**200** for reads, lists and actions; **201** when something is created.
Timestamps	RFC 3339 UTC with milliseconds, e.g. `2026-11-04T08:15:00.000Z`.
EPC	Upper-case hexadecimal words, e.g. `3034257BF7194E4000000001`.
Paging	Send `cursor` and `limit` (1 to 200, default 50) in the body; the response has `next_cursor` (null on the last page).
Auth	`Authorization: Bearer tgh_...` from `/handheld/login`. Only `/handheld/pair` and `/handheld/login` need no token. The token works only under `/handheld/`.
Idempotency	Count uploads carry a `batch_id`; bind and replace take an `Idempotency-Key` header. Repeating the same request returns the first result and changes nothing.
Errors	One envelope with a stable `code`, a message, a `request_id` and optional `details` (JSON pointers into the body). See Appendix A.

Roles

Role	Finder	Bind / replace	Counting	Admin
viewer	Finder	-	-	-
operator	Finder	-	Inventory count (create, upload reads, close own)	-
supervisor	Finder	Bind, replace	Inventory count and stock opname (create, review, resolve, reopen, close, cancel)	-
tenant_admin	Finder	Bind, replace	Everything above	Operators and handhelds (Site UI)

Typical flow

1 Pair (once)
Admin creates the handheld and gives a pairing code; the app calls `/handheld/pair` and keeps the device key.

2 Sign in
`/handheld/login` with the device key header, Nomor Operator and PIN returns a 12 h token.

3 Work
Finder, bind, count: every call carries the Bearer token.

4 Retry safely
After a network failure repeat the same `batch_id` or `Idempotency-Key`.

Counting flow

Create
`counts/create`
(inventory or opname)

Count
`counts/reads`
batches of up to 1,000 EPCs per area

Review
`counts/review`
computes missing, extra, unknown, short, over, area_conflict

Resolve
`counts/resolve`, or `reopen` to recount

Close
`counts/close`
freezes the report

Security notes

PINs are exactly 6 digits and are stored only as argon2id hashes. Five wrong PINs lock an operator for 15 minutes. A token is tied to one paired handheld, lives 12 hours, and is revoked at once when the admin disables the operator or the handheld. Tokens, PINs and device keys must never be logged by the app.

REFERENCE

Authentication

The handheld pairs once with a code from the site admin, then each operator signs in with Nomor Operator and PIN.

POST `/api/v1/handheld/pair`

Exchange a pairing code for a device key

Operation: `handheldPair` **Who:** public, rate limited **Auth:** none **Success:** 200

400 `pairing_invalid` when the code is wrong, expired or already used. The device key is shown once.

> The device key is shown once. Store it in the Android Keystore.

REQUEST BODY

Field	Type	Req.	Notes
`pairing_code`	string, min length 8, max length 8	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/pair \
  -H 'Content-Type: application/json' \
  -d '{"pairing_code":"K7M2QX9D"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`device_key`	string	yes	Shown once; the app keeps it in the Android Keystore and sends it as X-Handheld-Key.

EXAMPLE RESPONSE

```
{
  "device_key": "r7T2xN0cQe9LmV4bH1aZk8WjY3sDfG5uPoI6tXyC2nA"
}
```

ERRORS

400 validation_failed / pairing_invalid, **429** rate_limited, **501** not_implemented

---

POST `/api/v1/handheld/login`

Sign in with Nomor Operator and PIN

Operation: `handheldLogin` **Who:** public, needs the device key **Auth:** none **Success:** 200

401 `invalid_credentials` for a wrong PIN or an unknown operator (same answer). 403 `forbidden` for an unknown or disabled handheld. 423 `account_disabled`, 429 `rate_limited` after five wrong PINs (Retry-After).

HEADERS

Field	Type	Req.	Notes
`X-Handheld-Key`	string	yes	Device key issued by /handheld/pair.

REQUEST BODY

Field	Type	Req.	Notes
`operator_no`	string /^[0-9]{3,10}$/	yes	Nomor Operator, digits only, leading zeros kept.
`pin`	string /^[0-9]{6}$/	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/login \
  -H 'Content-Type: application/json' \
  -H 'X-Handheld-Key: <device key>' \
  -d '{"operator_no":"1042","pin":"482916"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`token`	string	yes	Opaque token tgh_..., valid for /handheld/ routes only, 12 h.
`expires_at`	string (date-time)	yes	RFC 3339 UTC with milliseconds.
`operator`	object	yes	
`operator.operator_no`	string /^[0-9]{3,10}$/	yes	Nomor Operator, digits only, leading zeros kept.
`operator.name`	string	yes	
`operator.role`	enum: viewer, operator, supervisor	yes	tenant_admin stays a web role.

EXAMPLE RESPONSE

```
{
  "token": "tgh_k3Q9v1mXz0aT7bW2yLr5uHn8cE4dJ6fPqGsA1oIxYw",
  "expires_at": "2026-11-04T20:15:00.000Z",
  "operator": {
    "operator_no": "1042",
    "name": "Budi Santoso",
    "role": "operator"
  }
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **423** account_disabled, **429** rate_limited, **501** not_implemented

---

POST `/api/v1/handheld/logout`

Revoke this token

Operation: `handheldLogout` **Who:** any operator **Auth:** Bearer token **Success:** 200

> No body. The token stops working at once.

REQUEST BODY

No body.

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/logout \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...'
```

RESPONSE 200

Empty body.

ERRORS

401 unauthenticated / invalid_credentials, **501** not_implemented

---

POST `/api/v1/handheld/me`

Operator, role, handheld and expiry

Operation: `handheldMe` **Who:** any operator **Auth:** Bearer token **Success:** 200

REQUEST BODY

No body.

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/me \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...'
```

RESPONSE 200

Field	Type	Req.	Notes
`operator`	object	yes	
`operator.operator_no`	string /^[0-9]{3,10}$/	yes	Nomor Operator, digits only, leading zeros kept.
`operator.name`	string	yes	
`operator.role`	enum: viewer, operator, supervisor	yes	tenant_admin stays a web role.
`handheld`	object	yes	
`handheld.id`	string (uuid)	yes	
`handheld.name`	string	yes	
`handheld.disabled`	boolean	yes	
`handheld.paired_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`handheld.last_seen_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`handheld.pairing_expires_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`expires_at`	string (date-time)	yes	RFC 3339 UTC with milliseconds.

EXAMPLE RESPONSE

```
{
  "operator": {
    "operator_no": "1042",
    "name": "Budi Santoso",
    "role": "operator"
  },
  "handheld": {
    "id": "0b9f5c4e-6d0e-4c1a-9a3e-2f6c1d7a8b01",
    "name": "HH-Gudang-01",
    "disabled": false,
    "paired_at": "2026-11-04T08:15:00.000Z",
    "last_seen_at": "2026-11-04T08:15:00.000Z"
  },
  "expires_at": "2026-11-04T20:15:00.000Z"
}
```

ERRORS

401 unauthenticated / invalid_credentials, **501** not_implemented

---

POST `/api/v1/handheld/pin`

Change the operator's own PIN

Operation: `handheldChangePin` **Who:** any operator **Auth:** Bearer token **Success:** 200

400 `validation_failed` for a wrong old PIN, a trivial new PIN or the operator number as PIN.

REQUEST BODY

Field	Type	Req.	Notes
`old_pin`	string /^[0-9]{6}$/	yes	
`new_pin`	string /^[0-9]{6}$/	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/pin \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"old_pin":"482916","new_pin":"730518"}'
```

RESPONSE 200

Empty body.

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **501** not_implemented

REFERENCE

Finder

Find items by SKU or by EPC. Read-only; every role can use it.

POST `/api/v1/handheld/finder/products`

Search products by SKU or name

Operation: `handheldFinderProducts` **Who:** viewer+ **Auth:** Bearer token **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`q`	string, max length 120	no	SKU or name.
`cursor`	string	no	
`limit`	integer, min 1, max 200	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/finder/products \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"q":"kemeja","limit":20}'
```

RESPONSE 200

Field	Type	Req.	Notes
`items`	array of object	yes	
`items[].product_id`	string (uuid)	yes	
`items[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`items[].name`	string	yes	
`items[].item_counts`	object	no	Items of the product by state (only the states that have items).
`next_cursor`	string or null	no	

EXAMPLE RESPONSE


```
{
  "items": [
    {
      "product_id": "9d1e7a33-2b54-4f0a-b6c8-1e5d3a7f9c03",
      "sku": "KMJ-BIRU-M",
      "name": "Kemeja Biru M",
      "item_counts": {
        "bound": 3,
        "encoded": 1
      }
    }
  ],
  "next_cursor": null
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **501** not_implemented

---

POST `/api/v1/handheld/finder/sku`

A product and its EPCs, paged, optional item state filter

Operation: `handheldFinderSku` **Who:** viewer+ **Auth:** Bearer token **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`state`	enum: reserved, encoded, void, bound, replaced, retired	no	
`cursor`	string	no	
`limit`	integer, min 1, max 200	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/finder/sku \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"sku":"KMJ-BIRU-M","state":"bound","limit":50}'
```

RESPONSE 200

Field	Type	Req.	Notes
`product`	object	yes	
`product.product_id`	string (uuid)	yes	
`product.sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`product.name`	string	yes	
`product.item_counts`	object	no	Items of the product by state (only the states that have items).
`items`	array of object	yes	
`items[].epc`	string /^([0-9A-F]{4}){1,31}$/	yes	
`items[].known`	boolean	yes	false when the EPC is neither an item nor a tag in the master.
`items[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	no	
`items[].name`	string	no	
`items[].serial`	integer (int64), min 1	no	
`items[].item_state`	enum: reserved, encoded, void, bound, replaced, retired	no	
`items[].tag_status`	enum: active, released, blocked	no	
`next_cursor`	string or null	no	

EXAMPLE RESPONSE


```
{
  "product": {
    "product_id": "9d1e7a33-2b54-4f0a-b6c8-1e5d3a7f9c03",
    "sku": "KMJ-BIRU-M",
    "name": "Kemeja Biru M",
    "item_counts": {
      "bound": 3,
      "encoded": 1
    }
  },
  "items": [
    {
      "epc": "3034257BF7194E4000000001",
      "known": true,
      "sku": "KMJ-BIRU-M",
      "name": "Kemeja Biru M",
      "serial": 1,
      "item_state": "bound",
      "tag_status": "active"
    },
    {
      "epc": "3034257BF7194E4000000002",
      "known": true,
      "sku": "KMJ-BIRU-M",
      "name": "Kemeja Biru M",
      "serial": 2,
      "item_state": "bound",
      "tag_status": "active"
    }
  ],
  "next_cursor": null
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **501** not_implemented

---

POST `/api/v1/handheld/finder/epc`

One EPC to product, item state and tag status

Operation: `handheldFinderEpc` **Who:** viewer+ **Auth:** Bearer token **Success:** 200

An EPC unknown to the master is answered 200 with `known: false`.

REQUEST BODY

Field	Type	Req.	Notes
`epc`	string /^([0-9A-F]{4}){1,31}$/	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/finder/epc \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"epc":"3034257BF7194E4000000001"}'
```

RESPONSE 200

Field	Type	Req.	Notes

`epc`	string /^([0-9A-F]{4}){1,31}$/	yes	
`known`	boolean	yes	false when the EPC is neither an item nor a tag in the master.
`sku`	string /^[A-Za-z0-9._/-]{1,40}$/	no	
`name`	string	no	
`serial`	integer (int64), min 1	no	
`item_state`	enum: reserved, encoded, void, bound, replaced, retired	no	
`tag_status`	enum: active, released, blocked	no	

EXAMPLE RESPONSE

```
{
  "epc": "3034257BF7194E4000000001",
  "known": true,
  "sku": "KMJ-BIRU-M",
  "name": "Kemeja Biru M",
  "serial": 1,
  "item_state": "bound",
  "tag_status": "active"
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **501** not_implemented

---

POST `/api/v1/handheld/finder/lookup`

Resolve a batch of up to 200 scanned EPCs

Operation: `handheldFinderLookup` **Who:** viewer+ **Auth:** Bearer token **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`epcs`	array of string /^([0-9A-F]{4}){1,31}$/	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/finder/lookup \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"epcs":["3034257BF7194E4000000001","E2801191A5030061FFFF739C"]}'
```

RESPONSE 200

Field	Type	Req.	Notes
`items`	array of object	yes	
`items[].epc`	string /^([0-9A-F]{4}){1,31}$/	yes	
`items[].known`	boolean	yes	false when the EPC is neither an item nor a tag in the master.
`items[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	no	

`items[].name`	string	no
`items[].serial`	integer (int64), min 1	no
`items[].item_state`	enum: reserved, encoded, void, bound, replaced, retired	no
`items[].tag_status`	enum: active, released, blocked	no

EXAMPLE RESPONSE


```
{
  "items": [
    {
      "epc": "3034257BF7194E4000000001",
      "known": true,
      "sku": "KMJ-BIRU-M",
      "name": "Kemeja Biru M",
      "serial": 1,
      "item_state": "bound",
      "tag_status": "active"
    },
    {
      "epc": "E2801191A5030061FFFF739C",
      "known": false
    }
  ]
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **501** not_implemented

REFERENCE

Binding and editing EPC

Bind a tag to a product, or replace a label. Supervisor and above. Send an Idempotency-Key so a retry is safe.

POST `/api/v1/handheld/items/bind`

Bind a tag to a product and serial

Operation: `handheldBindItem` **Who:** supervisor+, same rules as POST /items/bind, audited as item.bind **Auth:** Bearer token **Success:** 201

Same body and rules as `POST /items/bind`; docs/13 § 11. A retry with the same `Idempotency-Key` and body returns the first result.

HEADERS

Field	Type	Req.	Notes
`Idempotency-Key`	string, min length 8, max length 128	yes	A retry with the same key and body returns the first result (48 h).

REQUEST BODY

Field	Type	Req.	Notes
`epc`	string /^([0-9A-F]{4}){1,31}$/	yes	
`product_id`	string (uuid)	no	
`sku`	string /^[A-Za-z0-9._/-]{1,40}$/	no	
`gtin`	string /^[0-9]{14}$/	no	GTIN-14 (shorter GTINs left-padded with zeros); the check digit is validated.
`serial`	integer (int64), min 1	no	
`barcode_value`	string, min length 1, max length 80	no	
`tid`	string /^([0-9A-F]{4}){2,32}$/	no	TID as upper-case hex, whole 16-bit words (contract common.schema.json tid).

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/items/bind \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -H 'Idempotency-Key: bind-1042-20261104-0001' \
  -d '{"epc":"3034257BF7194E4000000001","sku":"KMJ-BIRU-M","serial":1}'
```

RESPONSE 201

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`product_id`	string (uuid)	yes	
`sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	

`serial`	integer (int64), min 1	yes	
`epc`	string /^([0-9A-F]{4}){1,31}$/	yes	
`epc_uri`	string	no	Pure identity URI for SGTIN-96 EPCs (urn:epc:id:sgtin:…); absent for private EPCs.
`barcode_value`	string	no	
`tid`	string /^([0-9A-F]{4}){2,32}$/	no	TID as upper-case hex, whole 16-bit words (contract common.schema.json tid).
`state`	enum: reserved, encoded, void, bound, replaced, retired	yes	
`origin`	enum: print, bind	yes	
`print_job_id`	string (uuid)	no	
`replaced_by`	string (uuid)	no	
`replaces`	string (uuid)	no	The item this one replaced.
`state_changed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`state_changed_by`	string	no	
`created_by`	string	yes	
`created_at`	string (date-time)	yes	RFC 3339 UTC with milliseconds.

EXAMPLE RESPONSE


```
{
  "id": "c4b8f2a1-7e39-4d5c-8f10-3a9b6e2d4c04",
  "product_id": "9d1e7a33-2b54-4f0a-b6c8-1e5d3a7f9c03",
  "sku": "KMJ-BIRU-M",
  "serial": 1,
  "epc": "3034257BF7194E4000000001",
  "state": "bound",
  "origin": "bind",
  "created_by": "operator:1042",
  "created_at": "2026-11-04T08:15:00.000Z"
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **409** conflict (see the code in the error body), **501** not_implemented

---

POST `/api/v1/handheld/items/replace`

Replace or edit the EPC of an item

Operation: `handheldReplaceItem` **Who:** supervisor+, same rules as POST /items/{id}/replace, audited as item.replace **Auth:** Bearer token **Success:** 201

Same rules as `POST /items/{id}/replace`, with `item_id` in the body. A retry with the same `Idempotency-Key` and body returns the first result.

HEADERS

Field	Type	Req.	Notes

`Idempotency-Key`	string, min length 8, max length 128	yes	A retry with the same key and body returns the first result (48 h).

REQUEST BODY

Field	Type	Req.	Notes
`comment`	string, min length 3, max length 500	yes	
`printer_id`	string (uuid)	no	
`template_id`	string (uuid)	no	Defaults to the product's template.
`spare_epc`	string /^([0-9A-F]{4}){1,31}$/	no	
`item_id`	string (uuid)	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/items/replace \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -H 'Idempotency-Key: replace-1042-20261104-0001' \
  -d '{"item_id":"c4b8f2a1-7e39-4d5c-8f10-3a9b6e2d4c04","spare_epc":"3034257BF7194E4000000002","commen
t":"Label rusak, ganti dengan tag cadangan"}'
```

RESPONSE 201

Field	Type	Req.	Notes
`item`	object	yes	
`item.id`	string (uuid)	yes	
`item.product_id`	string (uuid)	yes	
`item.sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`item.serial`	integer (int64), min 1	yes	
`item.epc`	string /^([0-9A-F]{4}){1,31}$/	yes	
`item.epc_uri`	string	no	Pure identity URI for SGTIN-96 EPCs (urn:epc:id:sgtin:…); absent for private EPCs.
`item.barcode_value`	string	no	
`item.tid`	string /^([0-9A-F]{4}){2,32}$/	no	TID as upper-case hex, whole 16-bit words (contract common.schema.json tid).
`item.state`	enum: reserved, encoded, void, bound, replaced, retired	yes	
`item.origin`	enum: print, bind	yes	
`item.print_job_id`	string (uuid)	no	
`item.replaced_by`	string (uuid)	no	
`item.replaces`	string (uuid)	no	The item this one replaced.
`item.state_changed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`item.state_changed_by`	string	no	

`item.created_by`	string	yes	
`item.created_at`	string (date-time)	yes	RFC 3339 UTC with milliseconds.
`print_job`	object	no	
`print_job.id`	string (uuid)	yes	
`print_job.number`	string /^PJ-[0-9]{4,}$/	yes	The readable job number: a per-site sequence, PJ- and at least four digits (PJ-0421); never reused, gaps are possible. What people read out; `id` stays the key.
`print_job.product_id`	string (uuid)	yes	
`print_job.sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`print_job.printer_id`	string (uuid)	yes	
`print_job.printer_name`	string	no	
`print_job.template_id`	string (uuid)	yes	
`print_job.template_version`	integer, min 1	yes	
`print_job.quantity`	integer, min 1, max 1000	yes	
`print_job.serial_first`	integer (int64)	yes	
`print_job.serial_last`	integer (int64)	yes	
`print_job.state`	enum: queued, running, paused, completed, cancelled	yes	
`print_job.pause_reason`	enum: printer_offline, printer_error, uncertain, restart, store_error, operator	no	
`print_job.encoded`	integer, min 0	yes	
`print_job.voided`	integer, min 0	yes	
`print_job.pending`	integer, min 0	yes	
`print_job.uncertain`	integer, min 0	yes	
`print_job.replaces_item_id`	string (uuid)	no	
`print_job.comment`	string	no	
`print_job.requested_by`	string	yes	
`print_job.created_at`	string (date-time)	yes	RFC 3339 UTC with milliseconds.
`print_job.started_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`print_job.finished_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.

EXAMPLE RESPONSE


```
{
  "item": {
    "id": "a1b2c3d4-e5f6-4a7b-8c9d-0e1f2a3b4c10",
    "product_id": "9d1e7a33-2b54-4f0a-b6c8-1e5d3a7f9c03",
    "sku": "KMJ-BIRU-M",
    "serial": 2,
    "epc": "3034257BF7194E4000000002",
    "state": "bound",
    "origin": "bind",
    "created_by": "operator:1042",
    "created_at": "2026-11-04T08:15:00.000Z",
    "replaces": "c4b8f2a1-7e39-4d5c-8f10-3a9b6e2d4c04"
  }
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **409** conflict (see the code in the error body), **501** not_implemented

REFERENCE

Inventory count and stock opname

An inventory count is an informal tally; a stock opname adds an expected-stock baseline, differences, review and a frozen report.

POST `/api/v1/handheld/counts/create`

Create a count session

Operation: `handheldCountCreate` **Who:** inventory: operator+ via counts.write; opname: supervisor+ via counts.review
Auth: Bearer token **Success:** 201

`kind` inventory needs the action `counts.write`, `opname` needs `counts.review`. For an opname with `expected_source` items the expected stock is snapshotted at creation (docs/16 § 6.3).

> kind opname needs counts.review (supervisor and above); inventory needs counts.write.

REQUEST BODY

Field	Type	Req.	Notes
`kind`	enum: inventory, opname	yes	
`scope`	object	no	
`scope.product_ids`	array of string (uuid)	no	
`scope.skus`	array of string /^[A-Za-z0-9._/-]{1,40}$/	no	
`expected_source`	enum: items, upload	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/create \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"kind":"opname","scope":{"skus":["KMJ-BIRU-M"]},"expected_source":"items"}'
```

RESPONSE 201

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`kind`	enum: inventory, opname	yes	
`number`	string	yes	Server generated
`status`	enum: open, review, closed, cancelled	yes	
`expected_source`	enum: items, upload	no	
`scope`	object	no	
`scope.product_ids`	array of string (uuid)	no	
`scope.skus`	array of string /^[A-Za-z0-9._/-]{1,40}$/	no	
`baseline_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`created_by`	string	yes	

`closed_by`	string	no	
`closed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`comment`	string	no	
`tally`	object	no	
`tally.distinct_epcs`	integer, min 0	yes	
`tally.by_sku`	array of object	yes	
`tally.by_sku[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`tally.by_sku[].counted`	integer, min 0	yes	
`tally.by_sku[].expected`	integer, min 0	no	
`tally.by_area`	array of object	yes	
`tally.by_area[].area`	string, max length 60	yes	
`tally.by_area[].counted`	integer, min 0	yes	

EXAMPLE RESPONSE


```
{
  "id": "e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05",
  "kind": "opname",
  "number": "SO-20261104-001",
  "status": "open",
  "expected_source": "items",
  "scope": {
    "skus": [
      "KMJ-BIRU-M"
    ]
  },
  "baseline_at": "2026-11-04T08:15:00.000Z",
  "created_by": "operator:2001",
  "tally": {
    "distinct_epcs": 2,
    "by_sku": [
      {
        "sku": "KMJ-BIRU-M",
        "counted": 2,
        "expected": 3
      }
    ],
    "by_area": [
      {
        "area": "Rak A3",
        "counted": 2
      }
    ]
  }
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **501** not_implemented

POST `/api/v1/handheld/counts/list`

List count sessions

Operation: `handheldCountList` **Who:** operator+, counts.write **Auth:** Bearer token **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`kind`	enum: inventory, opname	no	
`status`	enum: open, review, closed, cancelled	no	
`cursor`	string	no	
`limit`	integer, min 1, max 200	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/list \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"kind":"opname","status":"open","limit":20}'
```

RESPONSE 200

Field	Type	Req.	Notes
`items`	array of object	yes	
`items[].id`	string (uuid)	yes	
`items[].kind`	enum: inventory, opname	yes	
`items[].number`	string	yes	Server generated
`items[].status`	enum: open, review, closed, cancelled	yes	
`items[].expected_source`	enum: items, upload	no	
`items[].scope`	object	no	
`items[].scope.product_ids`	array of string (uuid)	no	
`items[].scope.skus`	array of string /^[A-Za-z0-9._/-]{1,40}$/	no	
`items[].baseline_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`items[].created_by`	string	yes	
`items[].closed_by`	string	no	
`items[].closed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`items[].comment`	string	no	
`items[].tally`	object	no	
`items[].tally.distinct_epcs`	integer, min 0	yes	
`items[].tally.by_sku`	array of object	yes	
`items[].tally.by_sku[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	

`items[].tally.by_sku[].counted`	integer, min 0	yes
`items[].tally.by_sku[].expected`	integer, min 0	no
`items[].tally.by_area`	array of object	yes
`items[].tally.by_area[].area`	string, max length 60	yes
`items[].tally.by_area[].counted`	integer, min 0	yes
`next_cursor`	string or null	no

EXAMPLE RESPONSE


```
{
  "items": [
    {
      "id": "e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05",
      "kind": "opname",
      "number": "SO-20261104-001",
      "status": "open",
      "expected_source": "items",
      "scope": {
        "skus": [
          "KMJ-BIRU-M"
        ]
      },
      "baseline_at": "2026-11-04T08:15:00.000Z",
      "created_by": "operator:2001",
      "tally": {
        "distinct_epcs": 2,
        "by_sku": [
          {
            "sku": "KMJ-BIRU-M",
            "counted": 2,
            "expected": 3
          }
        ],
        "by_area": [
          {
            "area": "Rak A3",
            "counted": 2
          }
        ]
      }
    }
  ],
  "next_cursor": null
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **501** not_implemented

---

POST `/api/v1/handheld/counts/get`

A count session with its running tally by SKU and area

Operation: `handheldCountGet` **Who:** operator+, counts.write **Auth:** Bearer token **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes

`id`	string (uuid)	yes

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/get \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`kind`	enum: inventory, opname	yes	
`number`	string	yes	Server generated
`status`	enum: open, review, closed, cancelled	yes	
`expected_source`	enum: items, upload	no	
`scope`	object	no	
`scope.product_ids`	array of string (uuid)	no	
`scope.skus`	array of string /^[A-Za-z0-9._/-]{1,40}$/	no	
`baseline_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`created_by`	string	yes	
`closed_by`	string	no	
`closed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`comment`	string	no	
`tally`	object	no	
`tally.distinct_epcs`	integer, min 0	yes	
`tally.by_sku`	array of object	yes	
`tally.by_sku[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`tally.by_sku[].counted`	integer, min 0	yes	
`tally.by_sku[].expected`	integer, min 0	no	
`tally.by_area`	array of object	yes	
`tally.by_area[].area`	string, max length 60	yes	
`tally.by_area[].counted`	integer, min 0	yes	

EXAMPLE RESPONSE


```
{
  "id": "e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05",
  "kind": "opname",
  "number": "SO-20261104-001",
  "status": "open",
  "expected_source": "items",
  "scope": {
    "skus": [
      "KMJ-BIRU-M"
    ]
  },
  "baseline_at": "2026-11-04T08:15:00.000Z",
  "created_by": "operator:2001",
  "tally": {
    "distinct_epcs": 2,
    "by_sku": [
      {
        "sku": "KMJ-BIRU-M",
        "counted": 2,
        "expected": 3
      }
    ],
    "by_area": [
      {
        "area": "Rak A3",
        "counted": 2
      }
    ]
  }
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **501** not_implemented

---

POST `/api/v1/handheld/counts/reads`

Upload reads; idempotent on batch_id

Operation: `handheldCountReads` **Who:** operator+, counts.write **Auth:** Bearer token **Success:** 200

Re-sending the same `batch_id` returns the first result and changes nothing. Up to 1000 reads and 256 KiB per batch. An EPC counts once per session.

> Send the same batch_id again after a network failure: replayed is true and nothing changes.

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`batch_id`	string (uuid)	yes	Idempotency key of this upload.
`area`	string, max length 60	no	Free-text area label, for example Rak A3.
`reads`	array of object	yes	
`reads[].epc`	string /^([0-9A-F]{4}){1,31}$/	yes	
`reads[].seen_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/reads \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05","batch_id":"6f2d8b4a-0c9e-4a1b-b5d7-8e3c1f5a9b07","area":"Rak A3","reads":[{"epc":"3034257BF7194E4000000001","seen_at":"2026-11-04T08:15:00.000Z"},{"epc":"E2801191A5030061FFFF739C","seen_at":"2026-11-04T08:15:00.000Z"}]}'
```

RESPONSE 200

Field	Type	Req.	Notes
`accepted`	integer, min 0	yes	EPCs new to the session.
`duplicates`	integer, min 0	yes	EPCs already counted.
`replayed`	boolean	yes	true when the batch_id was already processed (the first result is returned).
`tally`	object	yes	
`tally.distinct_epcs`	integer, min 0	yes	
`tally.by_sku`	array of object	yes	
`tally.by_sku[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`tally.by_sku[].counted`	integer, min 0	yes	
`tally.by_sku[].expected`	integer, min 0	no	
`tally.by_area`	array of object	yes	
`tally.by_area[].area`	string, max length 60	yes	
`tally.by_area[].counted`	integer, min 0	yes	

EXAMPLE RESPONSE

```
{
  "accepted": 2,
  "duplicates": 0,
  "replayed": false,
  "tally": {
    "distinct_epcs": 2,
    "by_sku": [
      {
        "sku": "KMJ-BIRU-M",
        "counted": 2,
        "expected": 3
      }
    ],
    "by_area": [
      {
        "area": "Rak A3",
        "counted": 2
      }
    ]
  }
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **409** conflict (see the code in the error body), **501** not_implemented

---

POST `/api/v1/handheld/counts/expected`

Upload expected lines for an opname while open

Operation: `handheldCountExpected` **Who:** supervisor+, counts.review **Auth:** Bearer token **Success:** 200

Replaces the snapshot of this opname. Lines are EPC-level (`epc`) or quantity-level (`sku` and `qty`); up to 5000 lines.

> Opname with expected_source upload only, while the session is open.

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`lines`	array of object	yes	
`lines[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	no	
`lines[].epc`	string /^([0-9A-F]{4}){1,31}$/	no	
`lines[].qty`	integer, min 0	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/expected \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05","lines":[{"sku":"KMJ-BIRU-M","qty":3}]}'
```

RESPONSE 200

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`kind`	enum: inventory, opname	yes	
`number`	string	yes	Server generated
`status`	enum: open, review, closed, cancelled	yes	
`expected_source`	enum: items, upload	no	
`scope`	object	no	
`scope.product_ids`	array of string (uuid)	no	
`scope.skus`	array of string /^[A-Za-z0-9._/-]{1,40}$/	no	
`baseline_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`created_by`	string	yes	
`closed_by`	string	no	
`closed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.

`comment`	string	no
`tally`	object	no
`tally.distinct_epcs`	integer, min 0	yes
`tally.by_sku`	array of object	yes
`tally.by_sku[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes
`tally.by_sku[].counted`	integer, min 0	yes
`tally.by_sku[].expected`	integer, min 0	no
`tally.by_area`	array of object	yes
`tally.by_area[].area`	string, max length 60	yes
`tally.by_area[].counted`	integer, min 0	yes

EXAMPLE RESPONSE


```
{
  "id": "e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05",
  "kind": "opname",
  "number": "SO-20261104-001",
  "status": "open",
  "expected_source": "items",
  "scope": {
    "skus": [
      "KMJ-BIRU-M"
    ]
  },
  "baseline_at": "2026-11-04T08:15:00.000Z",
  "created_by": "operator:2001",
  "tally": {
    "distinct_epcs": 2,
    "by_sku": [
      {
        "sku": "KMJ-BIRU-M",
        "counted": 2,
        "expected": 3
      }
    ],
    "by_area": [
      {
        "area": "Rak A3",
        "counted": 2
      }
    ]
  }
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **409** conflict (see the code in the error body), **501** not_implemented

---

POST `/api/v1/handheld/counts/review`

Stop counting and compute differences

Operation: `handheldCountReview` **Who:** supervisor+, counts.review **Auth:** Bearer token **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/review \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`session`	object	yes	
`session.id`	string (uuid)	yes	
`session.kind`	enum: inventory, opname	yes	
`session.number`	string	yes	Server generated
`session.status`	enum: open, review, closed, cancelled	yes	
`session.expected_source`	enum: items, upload	no	
`session.scope`	object	no	
`session.scope.product_ids`	array of string (uuid)	no	
`session.scope.skus`	array of string /^[A-Za-z0-9._/-]{1,40}$/	no	
`session.baseline_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`session.created_by`	string	yes	
`session.closed_by`	string	no	
`session.closed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`session.comment`	string	no	
`session.tally`	object	no	
`session.tally.distinct_epcs`	integer, min 0	yes	
`session.tally.by_sku`	array of object	yes	
`session.tally.by_sku[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`session.tally.by_sku[].counted`	integer, min 0	yes	
`session.tally.by_sku[].expected`	integer, min 0	no	
`session.tally.by_area`	array of object	yes	
`session.tally.by_area[].area`	string, max length 60	yes	
`session.tally.by_area[].counted`	integer, min 0	yes	

`by_kind`	object	yes	Differences by kind (only the kinds that occur).

EXAMPLE RESPONSE


```
{
  "session": {
    "id": "e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05",
    "kind": "opname",
    "number": "SO-20261104-001",
    "status": "review",
    "expected_source": "items",
    "scope": {
      "skus": [
        "KMJ-BIRU-M"
      ]
    },
    "baseline_at": "2026-11-04T08:15:00.000Z",
    "created_by": "operator:2001",
    "tally": {
      "distinct_epcs": 2,
      "by_sku": [
        {
          "sku": "KMJ-BIRU-M",
          "counted": 2,
          "expected": 3
        }
      ],
      "by_area": [
        {
          "area": "Rak A3",
          "counted": 2
        }
      ]
    }
  },
  "by_kind": {
    "missing": 1,
    "unknown": 1
  }
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **409** conflict (see the code in the error body), **501** not_implemented

---

POST `/api/v1/handheld/counts/differences`

Differences of a session, paged

Operation: `handheldCountDifferences` **Who:** supervisor+, counts.review **Auth:** Bearer token **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`kind`	enum: missing, extra, unknown, short, over, area_conflict	no	
`resolution`	enum: open, accepted, recount	no	
`cursor`	string	no	

`limit`	integer, min 1, max 200	no

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/differences \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05","kind":"missing","limit":50}'
```

RESPONSE 200

Field	Type	Req.	Notes
`items`	array of object	yes	
`items[].id`	string (uuid)	yes	
`items[].kind`	enum: missing, extra, unknown, short, over, area_conflict	yes	
`items[].epc`	string /^([0-9A-F]{4}){1,31}$/	no	
`items[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	no	
`items[].expected_qty`	integer, min 0	no	
`items[].counted_qty`	integer, min 0	no	
`items[].detail`	object	no	Tag status, item state, areas.
`items[].resolution`	enum: open, accepted, recount	yes	
`items[].resolved_by`	string	no	
`items[].comment`	string	no	
`next_cursor`	string or null	no	

EXAMPLE RESPONSE

```
{
  "items": [
    {
      "id": "1a3c5e7f-9b2d-4f6a-8c1e-3d5b7a9c2f06",
      "kind": "missing",
      "epc": "3034257BF7194E4000000002",
      "sku": "KMJ-BIRU-M",
      "resolution": "open"
    }
  ],
  "next_cursor": null
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **501** not_implemented

---

POST `/api/v1/handheld/counts/resolve`

Set the resolution and comment of one difference

Operation: `handheldCountResolve` **Who:** supervisor+, counts.review **Auth:** Bearer token **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`difference_id`	string (uuid)	yes	
`resolution`	enum: open, accepted, recount	yes	
`comment`	string, min length 1, max length 500	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/resolve \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05","difference_id":"1a3c5e7f-9b2d-4f6a-8c1e-3d5b7a9c2f0
6","resolution":"accepted","comment":"Barang sudah dikirim ke toko"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`kind`	enum: missing, extra, unknown, short, over, area_conflict	yes	
`epc`	string /^([0-9A-F]{4}){1,31}$/	no	
`sku`	string /^[A-Za-z0-9._/-]{1,40}$/	no	
`expected_qty`	integer, min 0	no	
`counted_qty`	integer, min 0	no	
`detail`	object	no	Tag status, item state, areas.
`resolution`	enum: open, accepted, recount	yes	
`resolved_by`	string	no	
`comment`	string	no	

EXAMPLE RESPONSE

```
{
  "id": "1a3c5e7f-9b2d-4f6a-8c1e-3d5b7a9c2f06",
  "kind": "missing",
  "epc": "3034257BF7194E4000000002",
  "sku": "KMJ-BIRU-M",
  "resolution": "accepted",
  "resolved_by": "operator:2001",
  "comment": "Barang sudah dikirim ke toko"
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **409** conflict (see the code in the error body), **501** not_implemented

POST `/api/v1/handheld/counts/reopen`

Move a session in review back to open

Operation: `handheldCountReopen` **Who:** supervisor+, counts.review **Auth:** Bearer token **Success:** 200

> Review back to open for a recount.

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/reopen \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`kind`	enum: inventory, opname	yes	
`number`	string	yes	Server generated
`status`	enum: open, review, closed, cancelled	yes	
`expected_source`	enum: items, upload	no	
`scope`	object	no	
`scope.product_ids`	array of string (uuid)	no	
`scope.skus`	array of string /^[A-Za-z0-9._/-]{1,40}$/	no	
`baseline_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`created_by`	string	yes	
`closed_by`	string	no	
`closed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`comment`	string	no	
`tally`	object	no	
`tally.distinct_epcs`	integer, min 0	yes	
`tally.by_sku`	array of object	yes	
`tally.by_sku[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`tally.by_sku[].counted`	integer, min 0	yes	
`tally.by_sku[].expected`	integer, min 0	no	
`tally.by_area`	array of object	yes	

`tally.by_area[].area`	string, max length 60	yes
`tally.by_area[].counted`	integer, min 0	yes

EXAMPLE RESPONSE


```
{
  "id": "e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05",
  "kind": "opname",
  "number": "SO-20261104-001",
  "status": "open",
  "expected_source": "items",
  "scope": {
    "skus": [
      "KMJ-BIRU-M"
    ]
  },
  "baseline_at": "2026-11-04T08:15:00.000Z",
  "created_by": "operator:2001",
  "tally": {
    "distinct_epcs": 2,
    "by_sku": [
      {
        "sku": "KMJ-BIRU-M",
        "counted": 2,
        "expected": 3
      }
    ],
    "by_area": [
      {
        "area": "Rak A3",
        "counted": 2
      }
    ]
  }
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **409** conflict (see the code in the error body), **501** not_implemented

---

POST `/api/v1/handheld/counts/close`

Close a session, comment required

Operation: `handheldCountClose` **Who:** supervisor+, counts.review; an inventory count may also be closed by its creator
Auth: Bearer token **Success:** 200

An opname must be in review. The report is frozen on close; unresolved differences stay listed as unresolved.

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`comment`	string, min length 3, max length 500	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/close \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05","comment":"Opname gudang A selesai"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`kind`	enum: inventory, opname	yes	
`number`	string	yes	Server generated
`status`	enum: open, review, closed, cancelled	yes	
`expected_source`	enum: items, upload	no	
`scope`	object	no	
`scope.product_ids`	array of string (uuid)	no	
`scope.skus`	array of string /^[A-Za-z0-9._/-]{1,40}$/	no	
`baseline_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`created_by`	string	yes	
`closed_by`	string	no	
`closed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`comment`	string	no	
`tally`	object	no	
`tally.distinct_epcs`	integer, min 0	yes	
`tally.by_sku`	array of object	yes	
`tally.by_sku[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`tally.by_sku[].counted`	integer, min 0	yes	
`tally.by_sku[].expected`	integer, min 0	no	
`tally.by_area`	array of object	yes	
`tally.by_area[].area`	string, max length 60	yes	
`tally.by_area[].counted`	integer, min 0	yes	

EXAMPLE RESPONSE


```
{
  "id": "e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05",
  "kind": "opname",
  "number": "SO-20261104-001",
  "status": "closed",
  "expected_source": "items",
  "scope": {
    "skus": [
      "KMJ-BIRU-M"
    ]
  },
  "baseline_at": "2026-11-04T08:15:00.000Z",
  "created_by": "operator:2001",
  "tally": {
    "distinct_epcs": 2,
    "by_sku": [
      {
        "sku": "KMJ-BIRU-M",
        "counted": 2,
        "expected": 3
      }
    ],
    "by_area": [
      {
        "area": "Rak A3",
        "counted": 2
      }
    ]
  },
  "closed_by": "operator:2001",
  "closed_at": "2026-11-04T08:15:00.000Z",
  "comment": "Opname gudang A selesai"
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **409** conflict (see the code in the error body), **501** not_implemented

---

POST `/api/v1/handheld/counts/cancel`

Cancel a session, comment required

Operation: `handheldCountCancel` **Who:** same roles as close **Auth:** Bearer token **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`comment`	string, min length 3, max length 500	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handheld/counts/cancel \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05","comment":"Salah gudang, dibatalkan"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`kind`	enum: inventory, opname	yes	
`number`	string	yes	Server generated
`status`	enum: open, review, closed, cancelled	yes	
`expected_source`	enum: items, upload	no	
`scope`	object	no	
`scope.product_ids`	array of string (uuid)	no	
`scope.skus`	array of string /^[A-Za-z0-9._/-]{1,40}$/	no	
`baseline_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`created_by`	string	yes	
`closed_by`	string	no	
`closed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`comment`	string	no	
`tally`	object	no	
`tally.distinct_epcs`	integer, min 0	yes	
`tally.by_sku`	array of object	yes	
`tally.by_sku[].sku`	string /^[A-Za-z0-9._/-]{1,40}$/	yes	
`tally.by_sku[].counted`	integer, min 0	yes	
`tally.by_sku[].expected`	integer, min 0	no	
`tally.by_area`	array of object	yes	
`tally.by_area[].area`	string, max length 60	yes	
`tally.by_area[].counted`	integer, min 0	yes	

EXAMPLE RESPONSE


```
{
  "id": "e7a1c5d9-1b28-4e6f-9c3a-5d8f2b4a6e05",
  "kind": "opname",
  "number": "SO-20261104-001",
  "status": "cancelled",
  "expected_source": "items",
  "scope": {
    "skus": [
      "KMJ-BIRU-M"
    ]
  },
  "baseline_at": "2026-11-04T08:15:00.000Z",
  "created_by": "operator:2001",
  "tally": {
    "distinct_epcs": 2,
    "by_sku": [
      {
        "sku": "KMJ-BIRU-M",
        "counted": 2,
        "expected": 3
      }
    ],
    "by_area": [
      {
        "area": "Rak A3",
        "counted": 2
      }
    ]
  },
  "closed_by": "operator:2001",
  "closed_at": "2026-11-04T08:15:00.000Z",
  "comment": "Salah gudang, dibatalkan"
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **409** conflict (see the code in the error body), **501** not_implemented

REFERENCE

Site admin: operators

Used from the Site UI session by a tenant_admin (not by the handheld).

POST `/api/v1/operators/list`

List operators

Operation: `listOperators` **Who:** tenant_admin, users.manage **Auth:** Site admin session cookie (tenant_admin) **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`cursor`	string	no	Opaque cursor from `next_cursor`.
`limit`	integer, min 1, max 200	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/operators/list \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"limit":50}'
```

RESPONSE 200

Field	Type	Req.	Notes
`items`	array of object	yes	
`items[].id`	string (uuid)	yes	
`items[].operator_no`	string /^[0-9]{3,10}$/	yes	Nomor Operator, digits only, leading zeros kept.
`items[].name`	string	yes	
`items[].role`	enum: viewer, operator, supervisor	yes	tenant_admin stays a web role.
`items[].disabled`	boolean	yes	
`items[].locked_until`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`items[].pin_changed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`items[].last_login_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`next_cursor`	string or null	no	

EXAMPLE RESPONSE

```
{
  "items": [
    {
      "id": "5c2a9e10-3f4b-4d7e-8a21-7b6e0c9d1f02",
      "operator_no": "1042",
      "name": "Budi Santoso",
      "role": "operator",
      "disabled": false,
      "pin_changed_at": "2026-11-04T08:15:00.000Z",
      "last_login_at": "2026-11-04T08:15:00.000Z"
    }
  ],
  "next_cursor": null
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **501** not_implemented

---

POST `/api/v1/operators/create`

Create an operator

Operation: `createOperator` **Who:** tenant_admin, users.manage **Auth:** Site admin session cookie (tenant_admin) **Success:** 201

Without `pin` the server generates one. The PIN is returned once and never again.

> Without a pin the server generates one. The PIN is shown once and is never stored in clear.

REQUEST BODY

Field	Type	Req.	Notes
`operator_no`	string /^[0-9]{3,10}$/	yes	Nomor Operator, digits only, leading zeros kept.
`name`	string, min length 1, max length 120	yes	
`role`	enum: viewer, operator, supervisor	yes	tenant_admin stays a web role.
`pin`	string /^[0-9]{6}$/	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/operators/create \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"operator_no":"1042","name":"Budi Santoso","role":"operator"}'
```

RESPONSE 201

Field	Type	Req.	Notes
`operator`	object	yes	
`operator.id`	string (uuid)	yes	

`operator.operator_no`	string /^[0-9]{3,10}$/	yes	Nomor Operator, digits only, leading zeros kept.
`operator.name`	string	yes	
`operator.role`	enum: viewer, operator, supervisor	yes	tenant_admin stays a web role.
`operator.disabled`	boolean	yes	
`operator.locked_until`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`operator.pin_changed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`operator.last_login_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`pin`	string /^[0-9]{6}$/	yes	

EXAMPLE RESPONSE


```
{
  "operator": {
    "id": "5c2a9e10-3f4b-4d7e-8a21-7b6e0c9d1f02",
    "operator_no": "1042",
    "name": "Budi Santoso",
    "role": "operator",
    "disabled": false,
    "pin_changed_at": "2026-11-04T08:15:00.000Z",
    "last_login_at": "2026-11-04T08:15:00.000Z"
  },
  "pin": "482916"
}
```


ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **409** conflict (see the code in the error body), **501** not_implemented

---

POST `/api/v1/operators/update`

Change name, role or disabled

Operation: `updateOperator` **Who:** tenant_admin, users.manage **Auth:** Site admin session cookie (tenant_admin) **Success:** 200

> Disabling an operator or changing the role ends that operator's handheld sessions at once.

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`name`	string, min length 1, max length 120	no	
`role`	enum: viewer, operator, supervisor	no	tenant_admin stays a web role.
`disabled`	boolean	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/operators/update \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"5c2a9e10-3f4b-4d7e-8a21-7b6e0c9d1f02","role":"supervisor"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`operator_no`	string /^[0-9]{3,10}$/	yes	Nomor Operator, digits only, leading zeros kept.
`name`	string	yes	
`role`	enum: viewer, operator, supervisor	yes	tenant_admin stays a web role.
`disabled`	boolean	yes	
`locked_until`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`pin_changed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`last_login_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.

EXAMPLE RESPONSE

```
{
  "id": "5c2a9e10-3f4b-4d7e-8a21-7b6e0c9d1f02",
  "operator_no": "1042",
  "name": "Budi Santoso",
  "role": "supervisor",
  "disabled": false,
  "pin_changed_at": "2026-11-04T08:15:00.000Z",
  "last_login_at": "2026-11-04T08:15:00.000Z"
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **501** not_implemented

---

POST `/api/v1/operators/reset-pin`

Set or generate a new PIN

Operation: `resetOperatorPin` **Who:** tenant_admin, users.manage **Auth:** Site admin session cookie (tenant_admin)
Success: 200

Clears the lockout and revokes tokens. The PIN is returned once.

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`pin`	string /^[0-9]{6}$/	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/operators/reset-pin \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"5c2a9e10-3f4b-4d7e-8a21-7b6e0c9d1f02"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`operator`	object	yes	
`operator.id`	string (uuid)	yes	
`operator.operator_no`	string /^[0-9]{3,10}$/	yes	Nomor Operator, digits only, leading zeros kept.
`operator.name`	string	yes	
`operator.role`	enum: viewer, operator, supervisor	yes	tenant_admin stays a web role.
`operator.disabled`	boolean	yes	
`operator.locked_until`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`operator.pin_changed_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`operator.last_login_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`pin`	string /^[0-9]{6}$/	yes	

EXAMPLE RESPONSE

```
{
  "operator": {
    "id": "5c2a9e10-3f4b-4d7e-8a21-7b6e0c9d1f02",
    "operator_no": "1042",
    "name": "Budi Santoso",
    "role": "operator",
    "disabled": false,
    "pin_changed_at": "2026-11-04T08:15:00.000Z",
    "last_login_at": "2026-11-04T08:15:00.000Z"
  },
  "pin": "730518"
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **501** not_implemented

REFERENCE

Site admin: handhelds

Used from the Site UI session by a tenant_admin (not by the handheld).

POST `/api/v1/handhelds/list`

List handhelds

Operation: `listHandhelds` **Who:** tenant_admin, users.manage **Auth:** Site admin session cookie (tenant_admin) **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`cursor`	string	no	Opaque cursor from `next_cursor`.
`limit`	integer, min 1, max 200	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handhelds/list \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"limit":50}'
```

RESPONSE 200

Field	Type	Req.	Notes
`items`	array of object	yes	
`items[].id`	string (uuid)	yes	
`items[].name`	string	yes	
`items[].disabled`	boolean	yes	
`items[].paired_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`items[].last_seen_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`items[].pairing_expires_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`next_cursor`	string or null	no	

EXAMPLE RESPONSE

```
{
  "items": [
    {
      "id": "0b9f5c4e-6d0e-4c1a-9a3e-2f6c1d7a8b01",
      "name": "HH-Gudang-01",
      "disabled": false,
      "paired_at": "2026-11-04T08:15:00.000Z",
      "last_seen_at": "2026-11-04T08:15:00.000Z"
    }
  ],
  "next_cursor": null
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **501** not_implemented

---

POST `/api/v1/handhelds/create`

Register a handheld and issue a pairing code

Operation: `createHandheld` **Who:** tenant_admin, users.manage **Auth:** Site admin session cookie (tenant_admin) **Success:** 201

The pairing code has 8 characters, lives 15 minutes and works once.

REQUEST BODY

Field	Type	Req.	Notes
`name`	string, min length 1, max length 120	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handhelds/create \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"name":"HH-Gudang-01"}'
```

RESPONSE 201

Field	Type	Req.	Notes
`handheld`	object	yes	
`handheld.id`	string (uuid)	yes	
`handheld.name`	string	yes	
`handheld.disabled`	boolean	yes	
`handheld.paired_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`handheld.last_seen_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`handheld.pairing_expires_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`pairing_code`	string, min length 8, max length 8	yes	Single use
`expires_at`	string (date-time)	yes	RFC 3339 UTC with milliseconds.

EXAMPLE RESPONSE

```
{
  "handheld": {
    "id": "0b9f5c4e-6d0e-4c1a-9a3e-2f6c1d7a8b01",
    "name": "HH-Gudang-01",
    "disabled": false,
    "pairing_expires_at": "2026-11-04T08:30:00.000Z"
  },
  "pairing_code": "K7M2QX9D",
  "expires_at": "2026-11-04T08:30:00.000Z"
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **409** conflict (see the code in the error body), **501** not_implemented

---

POST `/api/v1/handhelds/update`

Rename or disable a handheld

Operation: `updateHandheld` **Who:** tenant_admin, users.manage **Auth:** Site admin session cookie (tenant_admin) **Success:** 200

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`name`	string, min length 1, max length 120	no	
`disabled`	boolean	no	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handhelds/update \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"0b9f5c4e-6d0e-4c1a-9a3e-2f6c1d7a8b01","name":"HH-Gudang-01 (rusak)","disabled":true}'
```

RESPONSE 200

Field	Type	Req.	Notes
`id`	string (uuid)	yes	
`name`	string	yes	
`disabled`	boolean	yes	
`paired_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`last_seen_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`pairing_expires_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.

EXAMPLE RESPONSE

```
{
  "id": "0b9f5c4e-6d0e-4c1a-9a3e-2f6c1d7a8b01",
  "name": "HH-Gudang-01 (rusak)",
  "disabled": true,
  "paired_at": "2026-11-04T08:15:00.000Z",
  "last_seen_at": "2026-11-04T08:15:00.000Z"
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **409** conflict (see the code in the error body), **501** not_implemented

---

POST `/api/v1/handhelds/pairing-code`

Issue a new pairing code

Operation: `issuePairingCode` **Who:** tenant_admin, users.manage **Auth:** Site admin session cookie (tenant_admin)
Success: 200

REQUEST BODY

Field	Type	Req.	Notes
`id`	string (uuid)	yes	

EXAMPLE REQUEST

```
curl -X POST https://SITE-HOST/api/v1/handhelds/pairing-code \
  -H 'Content-Type: application/json' \
  -H 'Authorization: Bearer tgh_...' \
  -d '{"id":"0b9f5c4e-6d0e-4c1a-9a3e-2f6c1d7a8b01"}'
```

RESPONSE 200

Field	Type	Req.	Notes
`handheld`	object	yes	
`handheld.id`	string (uuid)	yes	
`handheld.name`	string	yes	
`handheld.disabled`	boolean	yes	
`handheld.paired_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`handheld.last_seen_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`handheld.pairing_expires_at`	string (date-time)	no	RFC 3339 UTC with milliseconds.
`pairing_code`	string, min length 8, max length 8	yes	Single use
`expires_at`	string (date-time)	yes	RFC 3339 UTC with milliseconds.

EXAMPLE RESPONSE

```
{
  "handheld": {
    "id": "0b9f5c4e-6d0e-4c1a-9a3e-2f6c1d7a8b01",
    "name": "HH-Gudang-01",
    "disabled": false,
    "paired_at": "2026-11-04T08:15:00.000Z",
    "last_seen_at": "2026-11-04T08:15:00.000Z"
  },
  "pairing_code": "R4TW8NZ2",
  "expires_at": "2026-11-04T08:30:00.000Z"
}
```

ERRORS

400 validation_failed / pairing_invalid, **401** unauthenticated / invalid_credentials, **403** forbidden (role or handheld not allowed), **404** not_found, **501** not_implemented

APPENDIX A

Error codes

Error body: `{"error": {"code": "validation_failed", "message": "...", "request_id": "...", "details": [{"path": "/pin", "code": "...", "message": "..."}]}}`. Give `request_id` to the site admin when reporting a problem. All Site API codes, including the ones this API uses:

Code	Status	Where
`validation_failed`	400	any write; details carry JSON pointers
`invalid_credentials`	401	POST /auth/login
`unauthenticated`	401	any call without a session
`forbidden`	403	role not allowed
`not_found`	404	gate, pass, document, user, outbox item
`document_exists`	409	POST /documents
`document_open`	409	POST /documents/{id}/bind; details[0].message = the open document number
`document_closed`	409	bind/close/PATCH of a non-open document
`document_readonly`	409	PATCH /documents/{id} of an ERP-sourced document (source erp:<adapter>): only bind and close apply (docs/12 S-11)
`no_alarm`	409	POST /passes/{id}/clear
`user_exists`	409	POST /users
`last_admin`	409	disable or demote the last tenant_admin
`already_done`	409	POST /erp/outbox/{id}/retry; POST /print-jobs/{id}/cancel|resume|pause of a completed or cancelled job
`product_exists`	409	POST /products, PATCH /products/{id}: SKU or GTIN taken; details[0].path names the field (ADR-028)
`item_exists`	409	POST /items/bind, /items/{id}/replace, /print-jobs: EPC, serial, barcode value or TID taken; details[0].path names the field, details[0].message the value (ADR-028)
`item_state`	409	an item, product or print job in a state that does not allow the action (replace a void item, change the GTIN of a product with items, print an inactive product, resume a running job, pause a job that is already paused, change numbering settings in use) (ADR-028)
`printer_unavailable`	409	POST /print-jobs, /items/{id}/replace: printer disabled or without RFID (ADR-028)
`printing_disabled`	409	POST /print-jobs (dry runs too), /print-jobs/{id}/resume without outcomes (with outcomes it resolves the labels and the job stays paused, R30-02), /items/{id}/replace, /items/bind, /tags/import with bind=true, /printers/{id}/test while features.printing is false (SITEGW_FEATURE_PRINTING=false or a private issuer conflict, features.printing_reason; docs/13 § 2, § 6.3)
`serial_exhausted`	409	POST /print-jobs: the product's serial range (SGTIN-96 < 2^38, private < 2^40) would be exceeded (ADR-028)
`count_state`	409	POST /handheld/counts/*: reads or an action on a session in the wrong state (docs/16 § 6.1)
`operator_exists`	409	POST /operators/create: operator number taken; details[0].path names the field
`handheld_exists`	409	POST /handhelds/create, /handhelds/update: name taken

`pairing_invalid`	400	POST /handheld/pair: code wrong, expired or already used
`account_disabled`	423	POST /auth/login
`rate_limited`	429	POST /auth/login (failed attempts per client address, IPv6 per /64, and per account); POST /auth/password (wrong current passwords per account)
`method_not_allowed`	405	a known path with another method
`not_implemented`	501	documents and ERP endpoints until milestone M2; tag printing endpoints on a gateway built without a printing store (ADR-028)
`internal`	500	any; a bug (docs/01 § 8), quote the X-Request-Id when reporting it

APPENDIX B

Test results

452 tests and checks passed and 1 failed across 17 groups. The one failure is outside this change: two TCC console tests that also fail on the untouched integration branch (reported to the main session).

Area	Command	Passed	Failed	Result
Build and vet	`go build ./... && go vet ./internal/sitegw/...`	1	0	PASS
Auth, PIN policy, lockout (unit)	`go test -race -count=1 -json ./internal/sitegw/auth/`	12	0	PASS
Site API handlers incl. handheld (unit)	`go test -race -count=1 -json ./internal/sitegw/api/`	144	0	PASS
Store conformance, fake store	`go test -race -count=1 -json ./internal/sitegw/store/...`	21	0	PASS
Store on PostgreSQL (integration)	`go test -race -count=1 -tags integration -json ./internal/sitegw/store/pg/`	82	0	PASS
API against PostgreSQL (integration)	`go test -race -count=1 -tags integration -json ./internal/sitegw/api/`	144	0	PASS
Count rules: differences (unit)	`go test -race -count=1 -json ./internal/sitegw/counting/`	3	0	PASS
Production wiring (integration)	`go test -race -count=1 -tags integration -json ./cmd/sitegw/`	37	0	PASS
Lint	`/root/taggiot-rfid/bin/golangci-lint run ./internal/sitegw/...`	1	0	PASS
Full gate run: Generated code in sync (make gen-check)	`scripts/ship.sh b2a61f70 --gates all`	1	0	PASS
Full gate run: Contract check (make contract-check)	`scripts/ship.sh b2a61f70 --gates all`	1	0	PASS
Full gate run: Whole-repo Go unit tests	`scripts/ship.sh b2a61f70 --gates all`	1	0	PASS
Full gate run: Whole-repo Go integration tests (one container-start timeout, rerun passed)	`scripts/ship.sh b2a61f70 --gates all`	1	0	PASS
Full gate run: Web: lint, format, typecheck, i18n	`scripts/ship.sh b2a61f70 --gates all`	1	0	PASS
Full gate run: Web: shared UI tests	`scripts/ship.sh b2a61f70 --gates all`	1	0	PASS
Full gate run: Web: Site UI tests (incl. roles matrix)	`scripts/ship.sh b2a61f70 --gates all`	1	0	PASS
Full gate run: Web: TCC console tests	`scripts/ship.sh b2a61f70 --gates all`	0	1	FAIL

Selected cases (35)

Test	What it proves	Result

`TestAuditOperator`	Audit helper records the operator and the handheld id	PASS
`TestCountsBatchReplay`	Same batch_id replays safely; another session is refused	PASS
`TestCountsInventoryFlow`	Inventory count: create, read, close	PASS
`TestCountsList`	Count list paging and filters	PASS
`TestCountsNeedAToken`	All count routes refuse anonymous and cookie callers	PASS
`TestCountsNotFound`	Unknown count session on every action	PASS
`TestCountsNumbersPerDay`	Numbers IC-/SO-yyyymmdd-nnn per local day	PASS
`TestCountsOpnameFlow`	Opname: snapshot, reads in two areas, review, resolve, reopen, review, close, audit	PASS
`TestCountsQuantityLines`	SKU quantity lines give short and over differences	PASS
`TestCountsRoles`	Operator may count; only supervisor and above run an opname	PASS
`TestCountsScope`	Scope filter limits the expected-stock snapshot	PASS
`TestCountsUploadSource`	Uploaded expected lines replace the snapshot	PASS
`TestCountsValidation`	40 invalid-input cases, each with the right error path	PASS
`TestCountsWrongState`	count_state 409 for every action from a wrong state	PASS
`TestEveryOperatorActionIsAudited`	Every write by an operator is audited with actor operator: <no>	PASS
`TestHandheldAdminRoutes`	Operator and handheld admin routes: roles, validation, conflicts	PASS
`TestHandheldBind`	Bind a tag: roles, validation, item_exists conflicts, audit	PASS
`TestHandheldChangePin`	PIN policy (no trivial PINs) and own PIN change	PASS
`TestHandheldChangePinIsAudited`	Own PIN change writes an audit row; a refused change writes none	PASS
`TestHandheldFinderEpc`	EPC to product, known and unknown EPCs	PASS
`TestHandheldFinderLookup`	Batch lookup keeps order, collapses duplicates, limit 200	PASS
`TestHandheldFinderNeedsToken`	Finder needs a valid token	PASS
`TestHandheldFinderProducts`	Product search with item counts	PASS
`TestHandheldFinderSku`	SKU to EPC list, cursor paging across pages, newest first	PASS
`TestHandheldIdempotency`	Same Idempotency-Key replays the first answer; one item, one audit row	PASS
`TestHandheldIdempotencyConcurrent`	Eight simultaneous requests with one key create one item	PASS
`TestHandheldIdempotencyKeyLength`	Key must be 8 to 128 characters (spec rule)	PASS
`TestHandheldLockout`	Five wrong PINs lock the operator; the lock survives a new server instance	PASS
`TestHandheldLogin`	Nomor Operator and PIN sign-in; 401, 403, 423, 429 cases; same 401 for unknown operator and wrong PIN	PASS
`TestHandheldPairing`	Pairing code works once, expires, and yields the device key	PASS

`TestHandheldReplace`	Replace a label with a spare EPC; old item replaced	PASS
`TestHandheldRoutesRefuseAnonymousAndCookie`	All 20+ bearer routes refuse anonymous and cookie callers	PASS
`TestHandheldSecretsNotRecorded`	PINs, tokens and device keys never appear in logs or audit rows	PASS
`TestHandheldSessions`	One live session per handheld; disabling an operator or handheld ends sessions at once	PASS
`TestHandheldTokenScope`	Bearer token works only under /handheld/; cookie sessions are refused there	PASS

> Run on 2026-11-04, commit `b2a61f70` . Full logs: `docs/handheld-api/test-output/` .